

Energy is one of the most important concepts in the world of science. It is present in the Universe **in a variety of forms**, including mechanical, chemical, electromagnetic, and nuclear energy. Even the inert mass of everyday matter contains a very large amount of energy. Although energy can be **transformed from one kind to another**, all observations and experiments to date suggest that the total amount of energy in the Universe never changes. That's also true for an isolated system, which is a collection of objects that can exchange energy with each other, but not with the rest of the Universe. If one form of energy in an isolated system decreases, then another form of energy in the system must increase. For example, if the system consists of a motor connected to a battery, the battery converts chemical energy to electrical energy and the motor converts electrical energy to mechanical energy. Understanding how energy changes from one form to another is essential in all the sciences one.

In this chapter **the focus** is mainly on mechanical energy, which is the sum of **kinetic energy**, the energy associated with motion, and **potential energy**—the energy associated with relative position. Using an energy approach to solve certain problems is **often much easier** than using forces and Newton's three laws. These two very different approaches are linked through the concept of work.

Chapter 5

Energy and Conservation Law

- 1. Work and Power**
- 2. Kinetic Energy & Work- Energy Principle**
- 3. Conservative and Nonconservative Forces**
- 4. Potential Energy**
- 5. Conservation of Energy**

5-1 Work and Power

(The study of **spatial process** about the relationship between **the force** and **movement**.)

The work done by a force $\vec{F}$ in displacing a body $\Delta\vec{s}$ as the scalar product of $\vec{F}$ and $\Delta\vec{s}$.

$$W = F\Delta s \cos\phi = \vec{F} \cdot \Delta\vec{S}$$

Work is a **scalar** quantity, no direction.

$$\left\{ \begin{array}{lll} 0 \leq \phi < \frac{\pi}{2} & dW > 0, & \text{positive work;} \\ \phi = \frac{\pi}{2} & dW = 0, & \text{zero work;} \\ \frac{\pi}{2} < \phi \leq \pi & dW < 0, & \text{negative work.} \end{array} \right.$$

Example 5-1(P127)

EXAMPLE 5.1 Sledding Through the Yukon

GOAL Apply the basic definitions of work done by a constant force.

PROBLEM An Eskimo returning from a successful fishing trip pulls a sled loaded with salmon. The total mass of the sled and salmon is 50.0 kg , and the Eskimo exerts a force of magnitude $1.20 \times 10^2\text{ N}$ on the sled by pulling on the rope. **(a)** How much work does he do on the sled if the rope is horizontal to the ground ($\theta = 0^\circ$ in Fig. 5.6) and he pulls the sled 5.00 m ? **(b)** How much work does he do on the sled if $\theta = 30.0^\circ$ and he pulls the sled the same distance? (Treat the sled as a point particle, so details such as the point of attachment of the rope make no difference.) **(c)** At a coordinate position of 12.4 m , the Eskimo lets up on the applied force. A friction force of 45.0 N between the ice and the sled brings the sled to rest at a coordinate position of 18.2 m . How much work does friction do on the sled?

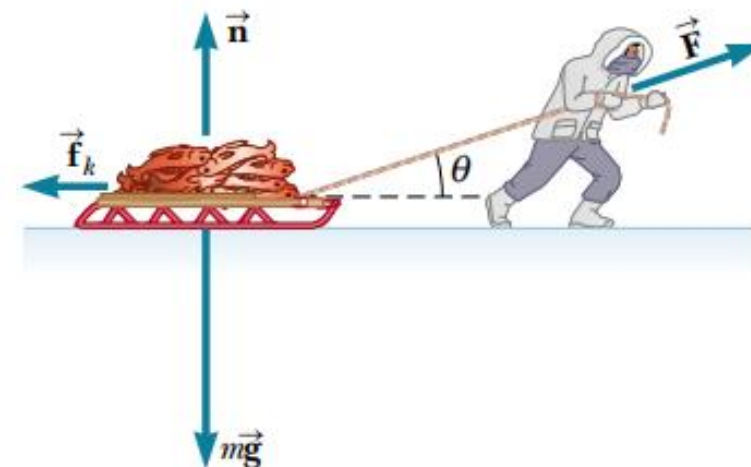


Figure 5.6 (Examples 5.1 and 5.2) An Eskimo pulling a sled with a rope at an angle θ to the horizontal.

Solution 5-1(P127)

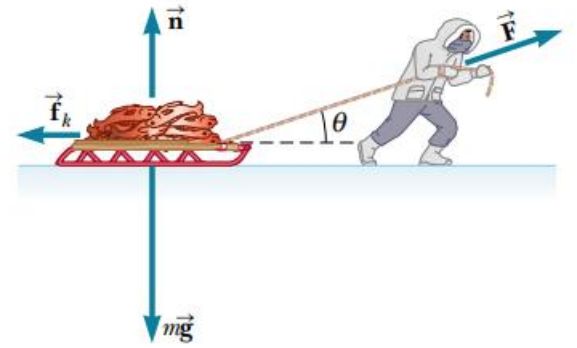


Figure 5.6 (Examples 5.1 and 5.2) An Eskimo pulling a sled with a rope at an angle θ to the horizontal.

SOLUTION

(a) Find the work done when the force is horizontal.

Use Equation 5.2, substituting the given values:

$$W = F_x \Delta x = (1.20 \times 10^2 \text{ N})(5.00 \text{ m}) = 6.00 \times 10^2 \text{ J}$$

(b) Find the work done when the force is exerted at a 30° angle.

Use Equation 5.3, again substituting the given values:

$$\begin{aligned} W &= (F \cos \theta) d = (1.20 \times 10^2 \text{ N})(\cos 30^\circ)(5.00 \text{ m}) \\ &= 5.20 \times 10^2 \text{ J} \end{aligned}$$

(c) How much work does a friction force of 45.0 N do on the sled as it travels from a coordinate position of 12.4 m to 18.2 m?

Use Equation 5.2, with F_x replaced by f_k :

$$W_{\text{fric}} = F_x \Delta x = f_k(x_f - x_i)$$

Substitute $f_k = -45.0 \text{ N}$ and the initial and final coordinate positions into x_i and x_f :

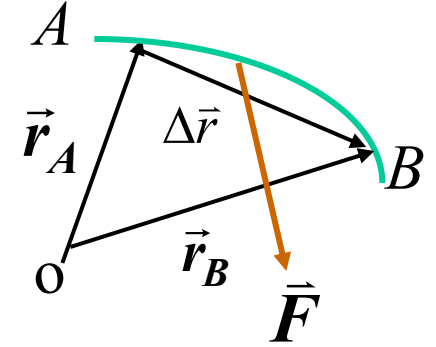
$$W_{\text{fric}} = (-45.0 \text{ N})(18.2 \text{ m} - 12.4 \text{ m}) = -261 \text{ J}$$

Explanations:

(1) If the force is **constant** in the whole path,

$$W = \vec{F} \cdot \int_A^B d\vec{r} = \vec{F} \cdot (\vec{r}_B - \vec{r}_A) = \vec{F} \cdot \Delta\vec{r}$$

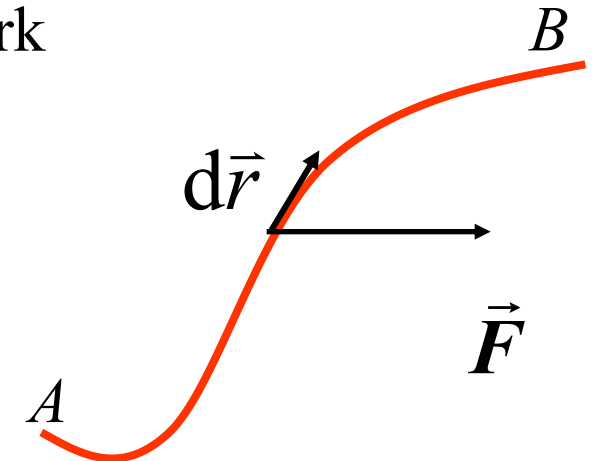
$$\Delta\vec{r} = \vec{r}_B - \vec{r}_A \text{ —displacement from } A \text{ to } B$$



(2) Work done by a **variable** force:

When a particle moves from A to B along a curve path, the total work done by the force equals the **integral** of $\vec{F}$ from A to B :

$$W = \int_A^B dW = \int_A^B \vec{F} \cdot d\vec{r}$$



Explanations:

(3) Work done by several forces: If $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \dots$

$$\begin{aligned} W &= \int \vec{F} \cdot d\vec{r} = \int \vec{F}_1 \cdot d\vec{r} + \int \vec{F}_2 \cdot d\vec{r} + \int \vec{F}_3 \cdot d\vec{r} + \dots \\ &= W_1 + W_2 + W_3 + \dots \end{aligned}$$

The work done by **several forces** = **algebraic sum** of the work done by **the individual force**.

(4) Special expression in scalar product: (In **cartesian coordinate system**)

$$\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}, \text{ and } d\vec{r} = dx \vec{i} + dy \vec{j} + dz \vec{k}.$$

$$\begin{aligned} W &= \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B (F_x dx + F_y dy + F_z dz) \\ &= \int_{x_i}^{x_f} F_x dx + \int_{y_i}^{y_f} F_y dy + \int_{z_i}^{z_f} F_z dz \end{aligned}$$

(5) Unit of work: In SI system, $1 \text{ N} \cdot \text{m} = 1 \text{ J (joule)} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$

Explanations:

(6) Instantaneous power:

The instantaneous power P is defined as the **rate** at which work is performed.

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

The SI unit of power is the **watt**, $1 \text{ W} = 1 \text{ J/s}$

$$1 \text{ horsepower} = 746 \text{ W}$$

5-2 Work-Kinetic Energy Theorem (P129)

A particle moves from a to b along an arbitrary path. $\vec{F}$ and $d\vec{s}$ may vary from point to point along the path.

$$\therefore dW = F \cos \phi \, ds$$

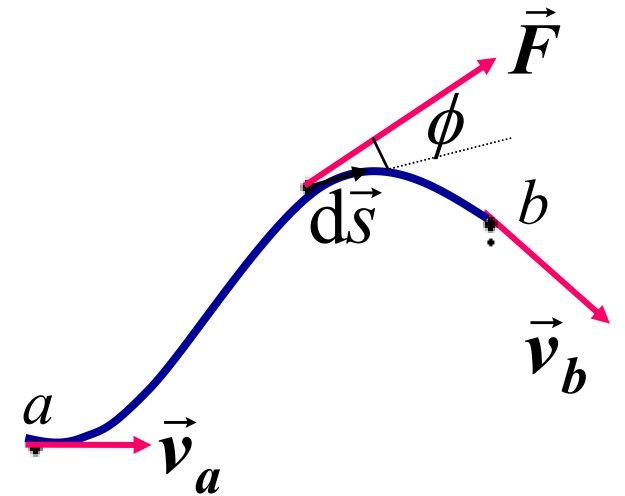
$$\text{and } F \cos \phi = F_t = ma_t = m \frac{dv}{dt}, \quad ds = v dt$$

(a_t is Tangential acceleration)

$$\therefore F \cos \phi \, ds = m \frac{dv}{dt} v dt = m v dv$$

$$\therefore W = m \int_{v_a}^{v_b} v dv = \frac{1}{2} m v_b^2 - \frac{1}{2} m v_a^2 = K_b - K_a$$

That is, $W_{\text{net}} = K_f - K_i = \Delta K$



Net work done on an object equals to the **change** in its **kinetic energy**.

5-2 Work-Kinetic Energy Theorem (P129)

Explanations:

$$W_{\text{net}} = K_{\text{f}} - K_{\text{i}} = \Delta K$$

(1) Work is the **measurements** for the **change in the kinetic energy (K)** of a body.

If $W > 0, K_{\text{f}} > K_{\text{i}}$ ———Body gains the kinetic energy.

$W < 0, K_{\text{f}} < K_{\text{i}}$ ———Body loses “ K ” & does work outside.

(2) Both work and kinetic energy are **scalars**. They have same units and dimensions; kinetic energy only depends on the speed of **initial** and **final** states, but work depends on the **real process**.

5-2 Work-Kinetic Energy Theorem (P129)

Explanations:

$$W_{\text{net}} = K_{\text{f}} - K_{\text{i}} = \Delta K$$

(3) For a system with more particles, work-kinetic energy theorem should be: The sum work done on a system by all external and internal forces equal to the change in kinetic energy of the system.

$$W_{\text{net}} = W_{\text{ext}} + W_{\text{int}} = \Delta K$$

(4) Both W and K depend on RF. Work-kinetic energy theorem is **only** hold for **inertial** RF.

Example (P130)

EXAMPLE 5.3 Collision Analysis

GOAL Apply the work–energy theorem with a known force.

PROBLEM The driver of a 1.00×10^3 kg car traveling on the interstate at 35.0 m/s (nearly 80.0 mph) slams on his brakes to avoid hitting a second vehicle in front of him, which had come to rest because of congestion ahead (Fig. 5.9). After the brakes are applied, a constant kinetic friction force of magnitude 8.00×10^3 N acts on the car. Ignore air resistance. **(a)** At what minimum distance should the brakes be applied to avoid a collision with the other vehicle? **(b)** If the distance between the vehicles is initially only 30.0 m, at what speed would the collision occur?

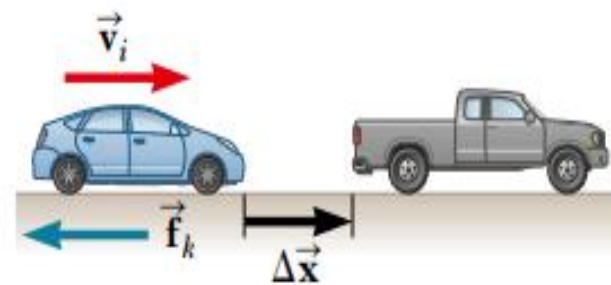


Figure 5.9 (Example 5.3) A braking vehicle just prior to an accident.

SOLUTION

(a) Find the minimum necessary stopping distance.

Apply the work–energy theorem to the car:

Substitute an expression for the frictional work and set $v_f = 0$:

Substitute $v_i = 35.0 \text{ m/s}$, $f_k = 8.00 \times 10^3 \text{ N}$, and $m = 1.00 \times 10^3 \text{ kg}$. Solve for Δx :

(b) At the given distance of 30.0 m, the car is too close to the other vehicle. Find the speed at impact.

Write down the work–energy theorem:

Multiply by $2/m$ and rearrange terms, solving for the final velocity v_f :

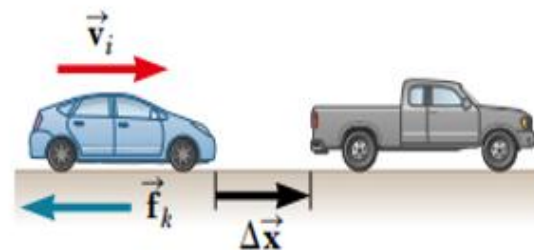


Figure 5.9 (Example 5.3) A braking vehicle just prior to an accident.

$$W_{\text{net}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$-f_k \Delta x = 0 - \frac{1}{2}mv_i^2$$

$$-(8.00 \times 10^3 \text{ N}) \Delta x = -\frac{1}{2}(1.00 \times 10^3 \text{ kg})(35.0 \text{ m/s})^2$$

$$\Delta x = 76.6 \text{ m}$$

$$W_{\text{net}} = W_{\text{fric}} = -f_k \Delta x = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$v_f^2 = v_i^2 - \frac{2}{m}f_k \Delta x$$

$$v_f^2 = (35.0 \text{ m/s})^2 - \left(\frac{2}{1.00 \times 10^3 \text{ kg}}\right)(8.00 \times 10^3 \text{ N})(30.0 \text{ m})$$

$$= 745 \text{ m}^2/\text{s}^2$$

$$v_f = 27.3 \text{ m/s}$$

5-3 Conservative and Nonconservative Forces (P131)

5-3-1 The features of the work done by universal gravitational force and the elastic force

1) Work done by gravity

Considering a body under $m\vec{g}$ near the surface of Earth, from a to b along the path of $\widehat{acb}$.

$$dW = \vec{F} \cdot d\vec{r}$$

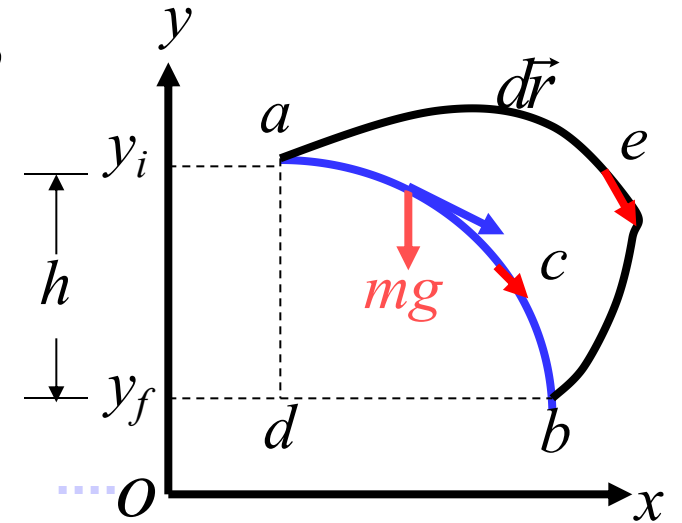
$$\vec{F} = -mg\vec{j}, \quad d\vec{r} = dx\vec{i} + dy\vec{j}$$

The work done by the gravity:

$$W = \int (-mg\vec{j}) \cdot d\vec{r} = \int (-mg\vec{j}) \cdot (dx\vec{i} + dy\vec{j}) = -mg \int_{y_i}^{y_f} dy$$



$$\therefore W_g = mgy_i - mgy_f = mgh$$



5-3-1 The features of the work done by universal gravitational force and the elastic force

The feature of the gravity:

★ $W_g = mgh$

◎ W_g on a particle depends *only* on the *initial and final position (or height)*, and does *not* depend on *the real path* taken by the particle.

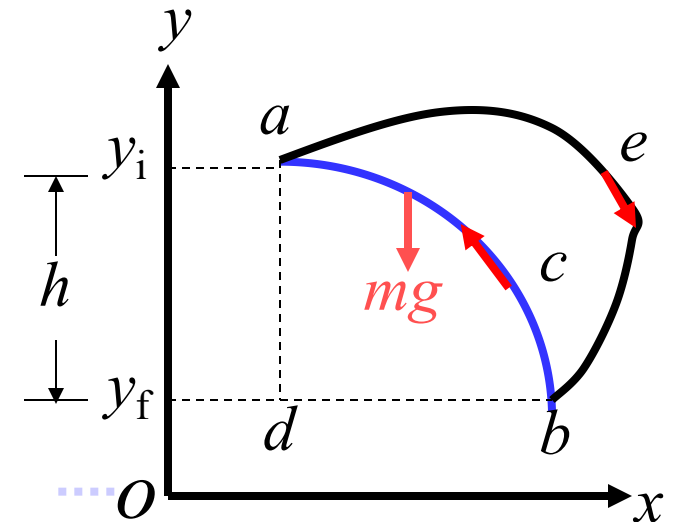
◎ W_g on a particle moving around a *closed* path=0.

$$W_{aebca} = W_{aeb} + W_{bca}$$

$$\therefore W_{aeb} = mg(y_i - y_f) = mgh,$$

$$W_{bca} = -mg(y_i - y_f) = -mgh$$

$$\therefore W_{aebca} = mgh - mgh = 0$$



5-3-1 The features of the work done by universal gravitational force and the elastic force

2) Work done by elastic force

Based on Hooke's law:

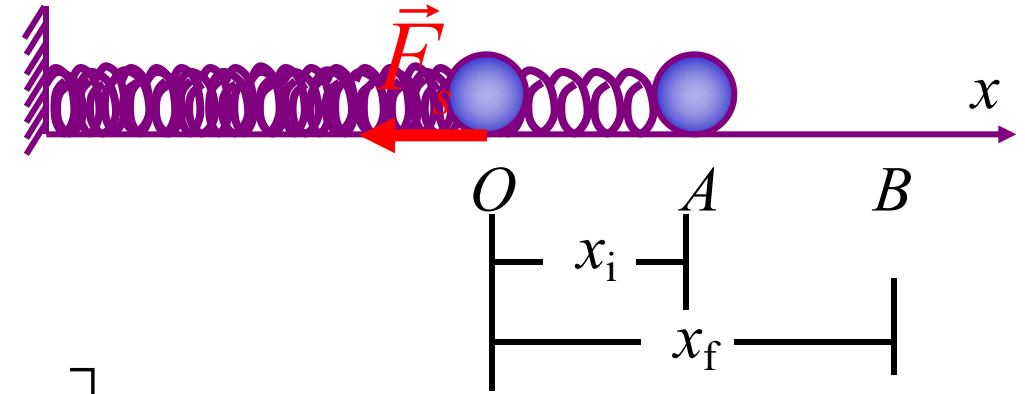
$$\vec{F}_s = -kx\vec{i} \quad k\text{-coefficients of elasticity}$$

$$W = \int \vec{F} \cdot d\vec{x} = \int_{x_i}^{x_f} -kx dx = -\left[\frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2 \right]$$

★

$$\therefore W_s = \frac{1}{2} kx_i^2 - \frac{1}{2} kx_f^2$$

$\vec{F}_s$ and $\vec{F}_g$ have the same features!



5-3 Conservative and Nonconservative Forces

5-3-2 Conservation force

Mathematically, the two common features of $\vec{F}_g$ and $\vec{F}_s$ can be written as:

Definition 1:
$$W = \oint_L \vec{F} \cdot d\vec{r} = 0$$

The net work done by conservative force on a particle moving around any closed path is zero.

Definition 2:
$$\int_A^B \vec{F} \cdot d\vec{s} = \text{constant}$$

Work done by a conservative force is recoverable or independence of paths .

Conservative Force

A force is conservative if the work it does moving an object between two points is the same no matter what path is taken.

Example

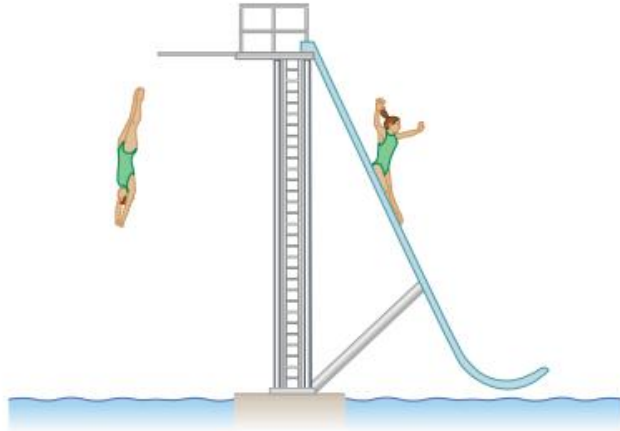


Figure 5.10 Because the gravity field is conservative, the diver regains as kinetic energy the work she did against gravity in climbing the ladder. Taking the frictionless slide gives the same result.

A diver climbing to the top of a 10-meter platform. The diver has to do work against gravity in making the climb. Once at the top, however, she can recover the work as kinetic energy by taking a dive. Her speed just before hitting the water will give her a kinetic energy equal to the work she did against gravity in climbing to the top of the platform, minus the effect of some nonconservative forces, such as air drag and internal muscular friction.

5-3 Conservative and Nonconservative Forces

5-3-3 Non-Conservation force

If work done by force is dependent on the path

$\oint_L \vec{F} \cdot d\vec{r} \neq 0$. The $\vec{F}$ is called **nonconservative force**.

A non-conservation force is also called **dissipative force**, such as a **friction** force.

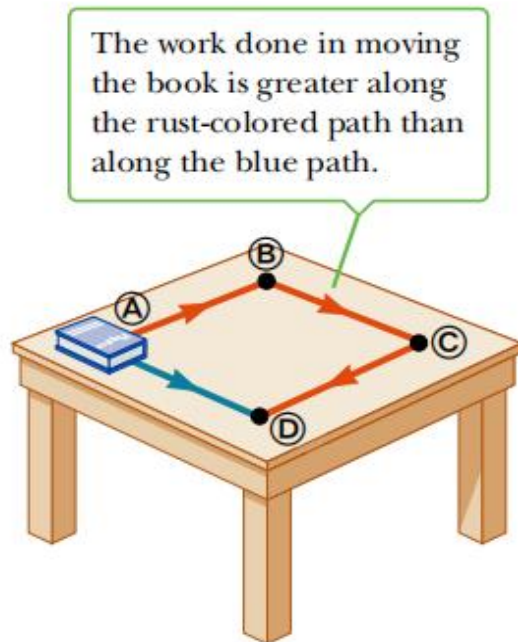
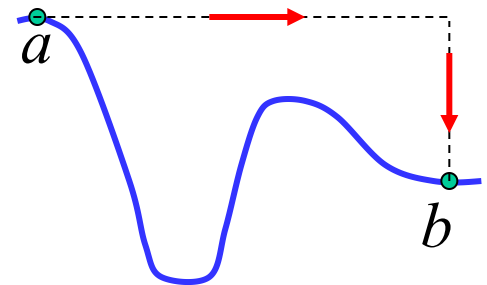
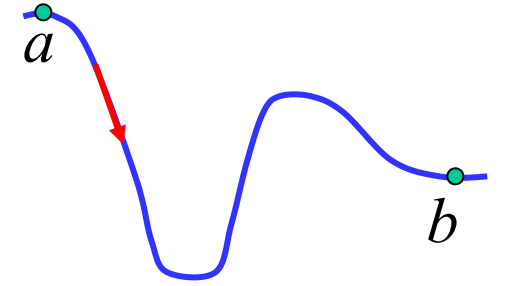


Figure 5.11 Because friction is a nonconservative force, a book pushed along the three segments A–B, B–C, and C–D requires three times the work as pushing the book directly from A to D.

Example

A 2.0kg block slides along a frictionless track from point a to b . It travels through a total distance of 2.0m along the track, and a net **vertical distance** of 0.80m. How much work is done on it by the gravitational force during the slide?

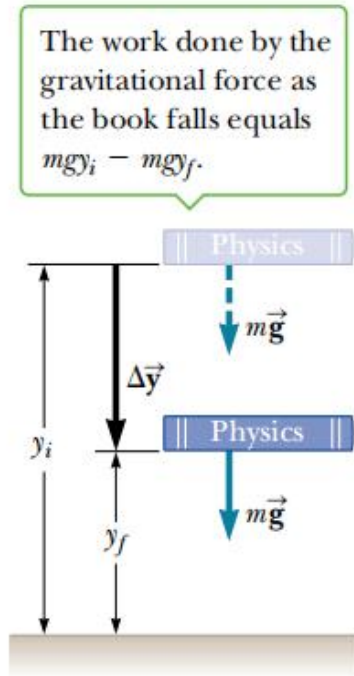


Solution:

$$\therefore W_v = mgd\cos 0 = 15.68\text{J}$$

$$\therefore W = W_h + W_v = 0 + 15.68\text{J} \approx 16\text{ J}$$

5-4 Potential Energy (P132)



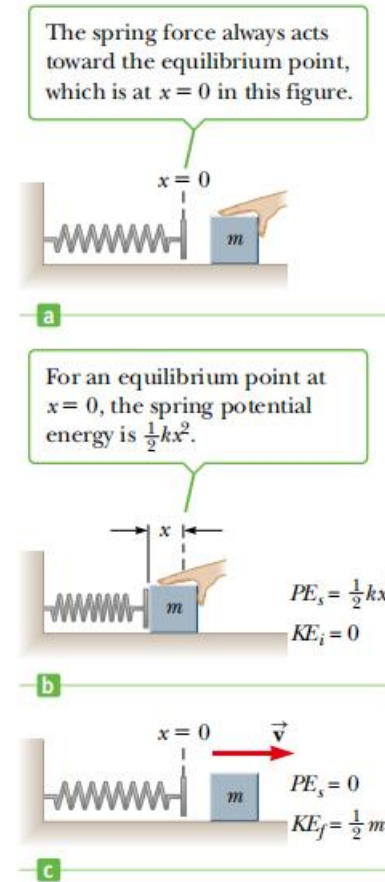
Active Figure 5.12 A book of mass m falls from a height y_i to a height y_f .

$$W_g = mgy_i - mgy_f$$

If we define a **work-function** — PE as $PE_g = mgy$

Then, $W_{\text{conser}} = -(PE_f - PE_i) = -\Delta PE$

The work done by conservative force = **minus** change of “PE” related to the conservative force



Active Figure 5.20

(a) A spring at equilibrium, neither compressed nor stretched. (b) A block of mass m on a frictionless surface is pushed against the spring. (c) When the block is released, the energy stored in the spring is transferred to the block in the form of kinetic energy.

$$W_s = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$$

$$PE_s = \frac{1}{2}kx^2$$

5-4 Potential Energy (P132)

Explanations:

☆ “PE” is the function of **states**.

☆ The relativity of potential energy:

$W_{ab} = PE_a - PE_b$ —— The difference of potential energy (“PE”) has absolute meaning, but to **assign** a particular PE value to an object **makes sense only if** the reference “PE” value is given.

Reference of “PE” is **arbitrary** to choose.

Definition of U :
$$PE_p = \int_{r_p}^{r_0} \vec{F}_{\text{con}} \cdot d\vec{r} \quad U_{r_0} = 0$$

Example

EXAMPLE 5.4 Wax Your Skis

GOAL Calculate the change in gravitational potential energy for different choices of reference level.

PROBLEM A 60.0-kg skier is at the top of a slope, as shown in Figure 5.14. At the initial point **A**, she is 10.0 m vertically above point **B**. (a) Setting the zero level for gravitational potential energy at **B**, find the gravitational potential energy of this system when the skier is at **A** and then at **B**. Finally, find the change in potential energy of the skier–Earth system as the skier goes from point **A** to point **B**. (b) Repeat this problem with the zero level at point **A**. (c) Repeat again, with the zero level 2.00 m higher than point **B**.

STRATEGY Follow the definition and be careful with signs. **A** is the initial point, with gravitational potential energy PE_i , and **B** is the final point, with gravitational potential energy PE_f . The location chosen for $y = 0$ is also the zero point for the potential energy, because $PE = mgy$.

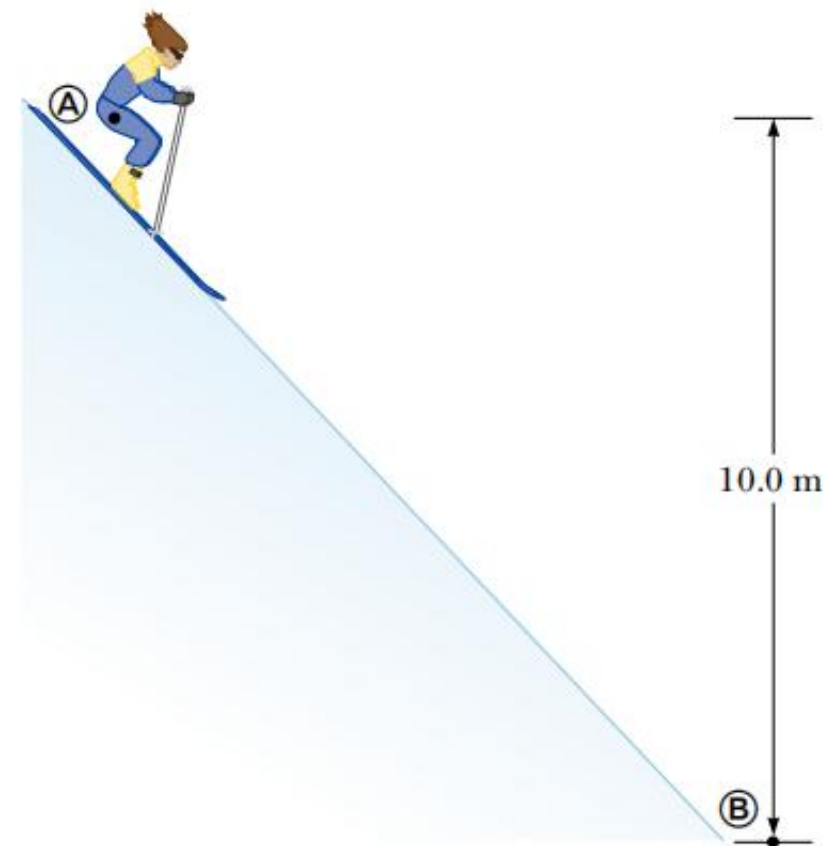


Figure 5.14 (Example 5.4)

SOLUTION

(a) Let $y = 0$ at ③. Calculate the potential energy at ① and at ③, and calculate the change in potential energy.

Find PE_i , the potential energy at ①, from Equation 5.11:

$PE_f = 0$ at ③ by choice. Find the difference in potential energy between ① and ③:

(b) Repeat the problem if $y = 0$ at ①, the new reference point, so that $PE = 0$ at ①.

Find PE_f , noting that point ③ is now at $y = -10.0$ m:

(c) Repeat the problem, if $y = 0$ two meters above ③.

Find PE_i , the potential energy at ①:

Find PE_f , the potential energy at ③:

Compute the change in potential energy:

$$PE_i = mgy_i = (60.0 \text{ kg})(9.80 \text{ m/s}^2)(10.0 \text{ m}) = 5.88 \times 10^3 \text{ J}$$

$$PE_f - PE_i = 0 - 5.88 \times 10^3 \text{ J} = -5.88 \times 10^3 \text{ J}$$

$$\begin{aligned} PE_f &= mgy_f = (60.0 \text{ kg})(9.80 \text{ m/s}^2)(-10.0 \text{ m}) \\ &= -5.88 \times 10^3 \text{ J} \end{aligned}$$

$$PE_f - PE_i = -5.88 \times 10^3 \text{ J} - 0 = -5.88 \times 10^3 \text{ J}$$

$$PE_i = mgy_i = (60.0 \text{ kg})(9.80 \text{ m/s}^2)(8.00 \text{ m}) = 4.70 \times 10^3 \text{ J}$$

$$\begin{aligned} PE_f &= mgy_f = (60.0 \text{ kg})(9.8 \text{ m/s}^2)(-2.00 \text{ m}) \\ &= -1.18 \times 10^3 \text{ J} \end{aligned}$$

$$\begin{aligned} PE_f - PE_i &= -1.18 \times 10^3 \text{ J} - 4.70 \times 10^3 \text{ J} \\ &= -5.88 \times 10^3 \text{ J} \end{aligned}$$

REMARKS These calculations show that the change in the gravitational potential energy when the skier goes from the top of the slope to the bottom is $-5.88 \times 10^3 \text{ J}$, regardless of the zero level selected.

Example

EXAMPLE 5.9 A Horizontal Spring

GOAL Use conservation of energy to calculate the speed of a block on a horizontal spring with and without friction.

PROBLEM A block with mass of 5.00 kg is attached to a horizontal spring with spring constant $k = 4.00 \times 10^2 \text{ N/m}$, as in Figure 5.21. The surface the block rests upon is frictionless. If the block is pulled out to $x_i = 0.0500 \text{ m}$ and released, (a) find the speed of the block when it first reaches the equilibrium point, (b) find the speed when $x = 0.0250 \text{ m}$,

STRATEGY In parts (a) and (b) there are no nonconservative forces, so conservation of energy, Equation 5.19, can be applied. In part (c) the definition of work and the work-energy theorem are needed to deal with the loss of mechanical energy due to friction.

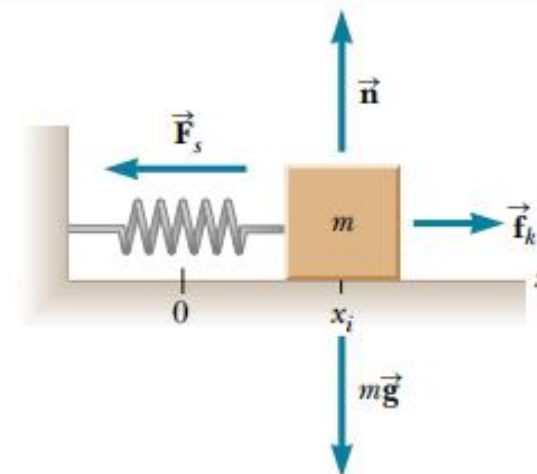


Figure 5.21 (Example 5.9) A mass attached to a spring.

SOLUTION

(a) Find the speed of the block at equilibrium point.

Start with Equation 5.19:

Substitute expressions for the block's kinetic energy and the potential energy, and set the gravity terms to zero:

Substitute $v_i = 0$, $x_f = 0$, and multiply by $2/m$:

Solve for v_f and substitute the given values:

(b) Find the speed of the block at the halfway point.

Set $v_i = 0$ in Equation (1) and multiply by $2/m$:

Solve for v_f and substitute the given values:

$$(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$$

$$(1) \quad \frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

$$\frac{k}{m}x_i^2 = v_f^2$$

$$\begin{aligned} v_f &= \sqrt{\frac{k}{m}x_i^2} = \sqrt{\frac{4.00 \times 10^2 \text{ N/m}}{5.00 \text{ kg}}}(0.0500 \text{ m}) \\ &= 0.447 \text{ m/s} \end{aligned}$$

$$\frac{kx_i^2}{m} = v_f^2 + \frac{kx_f^2}{m}$$

$$\begin{aligned} v_f &= \sqrt{\frac{k}{m}(x_i^2 - x_f^2)} \\ &= \sqrt{\frac{4.00 \times 10^2 \text{ N/m}}{5.00 \text{ kg}}[(0.050 \text{ m})^2 - (0.025 \text{ m})^2]} \\ &= 0.387 \text{ m/s} \end{aligned}$$

5-5 Mechanical Energy and Its Conservation (P145)

In physics, energy have different forms. One form of energy can transfer to another form, but it can not magically appear or disappear. They obey a law called the law of **Conservation of Energy**

5-5-1 Work-energy principle

The sum work done on a system by all **external** and **internal** forces equal to **the change in kinetic energy** of the system.

$$W_{\text{net}} = \Delta K \quad (1)$$

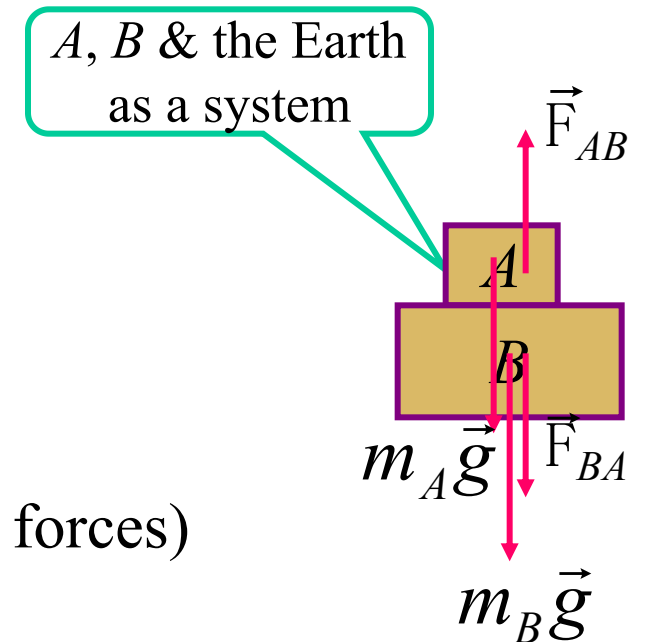
where $\sum_i W_{\text{ext}} = W_{\text{ext}}$

and $\sum_i W_{\text{int}} = W_{\text{ci}} + W_{\text{nci}}$

(W_{int} is the work done by conservative & **non**conservative **internal** forces)

Internal Force: $\vec{F}_{AB}, \vec{F}_{BA}, m_A \vec{g}, m_B \vec{g}$

External Force: 0



$$W_{\text{ext}} + W_{c_i} + W_{nc_i} = K_f - K_i \quad (2)$$

$$\because W_{c_i} = -(U_f - U_i) \quad (3)$$

(the work done by **conservative force** = minus change of potential energy related to the conservative force)

$$W_{\text{ext}} + W_{nc_i} = (K_f + U_f) - (K_i + U_i) = \Delta E_{\text{mec}} \quad (4)$$

(the work done by **external forces** and **nonconservative internal forces** = change of **mechanical energy** of a system)

or $\boxed{W_{\text{noncon}} = \Delta E_{\text{mec}}}$ —The theorem of work-energy

(the work done by **nonconservative forces** = change of **mechanical energy** of a system)

$$E_{\text{mec}} = K + U \quad \text{—mechanical energy of a system.}$$

Kinetic Energy

Potential Energy

5-5-2 The law of conservation of mechanical energy

$$W_{\text{ext}} + W_{nc_i} = (K_f + U_f) - (K_i + U_i) = \Delta E_{\text{mec}}$$

(the work done by **external forces** and **nonconservative internal forces**= change of **mechanical energy** of a system)

$$\text{when } W_{\text{ext}} + W_{nc_i} = 0 \quad \Rightarrow \quad K_f + U_f = K_i + U_i$$

If **only** conservative forces do work, mechanical energy **conserves**

—— both kinetic energy and potential energy transfer each other.

Example

EXAMPLE 5.10 Circus Acrobat

GOAL Use conservation of mechanical energy to solve a one-dimensional problem involving gravitational potential energy and spring potential energy.

PROBLEM A 50.0-kg circus acrobat drops from a height of 2.00 meters straight down onto a springboard with a force constant of $8.00 \times 10^3 \text{ N/m}$, as in Figure 5.22. By what maximum distance does she compress the spring?

STRATEGY Nonconservative forces are absent, so conservation of mechanical energy can be applied. At the two points of interest, the acrobat's initial position and the point of maximum spring compression, her velocity is zero, so the kinetic energy terms will be zero. Choose $y = 0$ as the point of maximum compression, so the final gravitational potential energy is zero. This choice also means that the initial position of the acrobat is $y_i = h + d$, where h is the acrobat's initial height above the platform and d is the spring's maximum compression.

SOLUTION

Use conservation of mechanical energy:

The only nonzero terms are the initial gravitational potential energy and the final spring potential energy.

Substitute the given quantities and rearrange the equation into standard quadratic form:

Solve with the quadratic formula (Equation A.8):

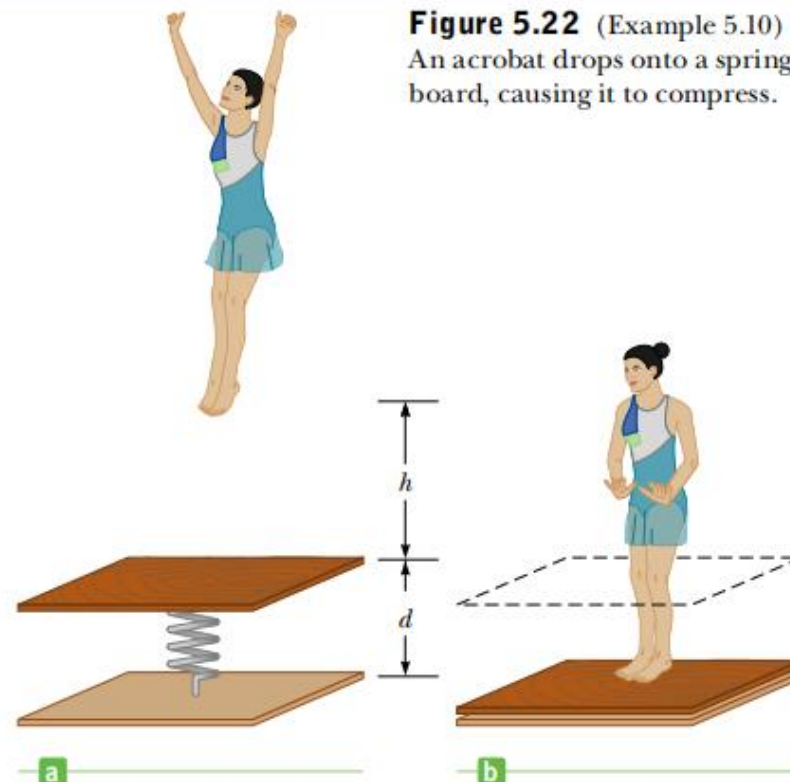


Figure 5.22 (Example 5.10) An acrobat drops onto a springboard, causing it to compress.

$$(1) \quad (KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$$

$$0 + mg(h + d) + 0 = 0 + 0 + \frac{1}{2}kd^2$$

$$mg(h + d) = \frac{1}{2}kd^2$$

$$(50.0 \text{ kg})(9.80 \text{ m/s}^2)(2.00 \text{ m} + d) = \frac{1}{2}(8.00 \times 10^3 \text{ N/m})d^2$$

$$d^2 - (0.123 \text{ m})d - 0.245 \text{ m}^2 = 0$$

$$d = 0.560 \text{ m}$$